

Computer-Aided Design (CAD)

This is the use of computer software to assist with the design, layout, and technical documentation of products and structures using 2D and 3D Modelling. CAD enables engineers, architects, city planners, artist, and designers to generate two-dimensional (2D) or three-dimensional (3D) models of an object or system of objects and view those models under a variety of different parameters to simulate and test real-world product conditions.

- CAD stands for Computer Aided Design. It is used to design, develop and optimize products.
- CAD is extensively used in the design of tools and equipment required in the manufacturing process as well as in the construction domain.
- CAD enables design engineers to layout and to develop their work on a computer screen, print and save it for future editing.

Gone are the days when engineers and architects used pencils, paper, T squares and triangles to compile a technical drawing. CAD has rendered obsolete the need to refill your mechanical pencil when you run out of lead, use an eraser when you make a mistake or walk across the room to share the drawing with another person. CAD files can be copied and distributed easily using the Internet, making it very simple to collaborate.

CAD began in **1957** when **Dr. Patrick J. Hanratty** developed **Pronto**. Pronto was the world's first commercial numerical-control programming system. In fact, designers today say that Dr. Hanratty was the '*father of CAD/CAM.*' Then **Ivan Sutherland**, of MIT's Lincoln Laboratory, designed the **sketchpad** in 1960.

A special printer or plotter is usually required for printing professional design renderings. CAD programs use either vector-based graphics or raster graphics that show how an object will look.

Today, the use of CAD has permeated almost all industries. From aerospace, shipbuilding, electronics, designs, engineering and manufacturing, CAD is used in all industry verticals. Since CAD encourages creativity and speeds up productivity, it is becoming more and more useful as an important tool for visualization before actually implementing a manufacturing process.

Examples of CAD software/Applications

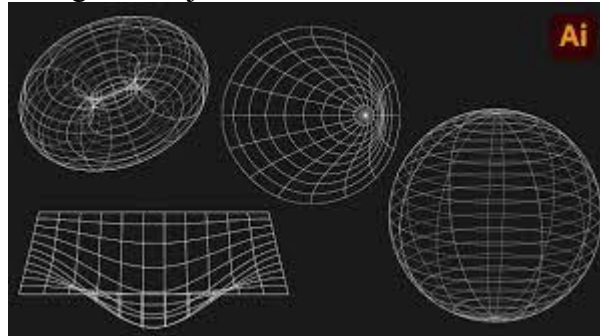
- **AutoCAD**, **3ds Max** formally 3D Studio, **Inventor**, **Fusion360** and **Maya** — it is the most popular commercial CAD software titles published by Autodesk Inc.
- **Blender** — an open-source CAD, animation, and image processing application with an active community of users.
- **OpenSCAD** is a software that is used for creating solid 3D CAD models. It is completely free and open source, and is available for GNU/Linux, Microsoft Windows and Mac OS.
- **SketchUp** — a proprietary CAD application that runs in a web browser, formerly developed by Google.
- **Solid Edge** — Originally released in 1996 by Intergraph and later sold to and distributed by Siemens, this modeling technique mixes parametric and direct modelling and offers a high-level control with the help of history-based modeling.

- **SmartDraw** — Diagramming tool used to flowcharts, organization charts, mind maps, project charts, networks and engineering diagrams, and other business visuals
- **SolidWorks** — The software includes features such as design validation tools, or reverse engineering. It tends to be used for industrial objects.
- **TinkerCAD, FreeCAD, BlocksCAD** are all beginners level CAD software's.

Types of CAD Software Systems

A broad classification of CAD is:

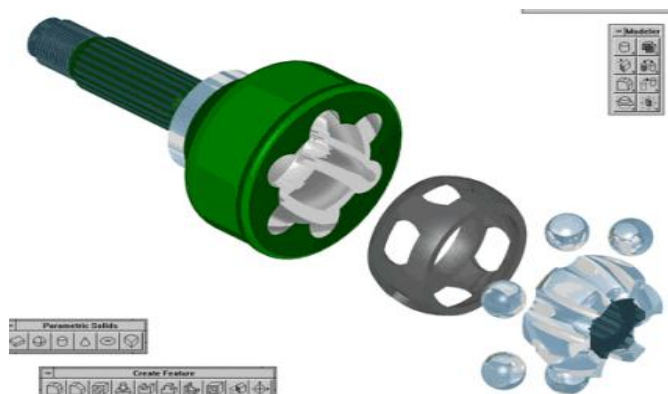
- **2D CAD**
- **3D CAD (which is further subdivided into 3)**
 - 3D wireframes – This model is an edge or skeletal representation of a real-life object. It consists of lines, points, arcs, circles, and other curves that define the center lines or edges of objects.



- Surface modeling focuses more on the external (skin) aspects of an object. It is essentially describing the 3D surface boundaries of the object.



- Solid Modelling – process of constructing a 3D model through a series of additive and subtractive operations. modeling of 3D solid objects. The objective is to ensure that every surface is geometrically correct.



Uses of Computer Aided Design (CAD)

- It is used in Creating and releasing 2D and 3D drawings Models of a physical objects.
- CAD is Interfacing with analysis, marketing, manufacturing, and end-user personnel.

- It allows the object to be viewed from any angle, even from the inside looking out.
- CAD facilitates manufacturing process by transferring detailed information about a product in an automated form that can be universally interpreted by trained personnel.

Benefits of Computer Aided Designs

- 'Editing' is a fast process as compared to manual method.
- CAD helps in designing first time right products and reducing overall cost.
- CAD is widely used from conceptual design and layout of products to definition of manufacturing of components.
- CAD reduces design time by allowing precise simulation rather than build and test physical prototypes.
- It can be Integrated with CAM (Computer Aided Manufacturing) to streamline the product development.
- CAD is currently widely used for industrial products, animated movies and other applications.

Advantages of CAD

- Increased productivity.
- Improved precision.
- Detailed engineering of 2D and 3D modelling.
- Better digital visualization.
- Faster collaboration.
- verification of design before actually building it.
- Reduced product design timelines
- Reduced Cost.

Disadvantage of CAD

- High initial Investment.
- Licencing fee.
- Design Work can be lost either by viruses or Pc breakdown
- Time and cost of training

CAD is used in the following industries but not limited to;

1. Production Design and Manufacturing.
2. Architectural, Civil, Building, Mechanical Engineering.
3. Construction
4. aerospace industries
5. Automotive industries
6. Gaming Industries and Virtual Reality
7. Manufacturing
8. Shipbuilding
9. Animations